

TMS TASKS

Task #1

1.1 Shipwrecks

- ~~Use~~ Use visual clues to identify shipwreck

- determine ship type

- determine length

- determine cargo

→ ~~C&C~~ C&C

- Remove cover & determine cargo

- Replace cover

→ TMS ~~C&C~~ C&C

- identify ship wreck

→ C&C

- Create 360 photosphere

1.2 Smart Buoy

- Replace damaged thermistor

→ TMS

- Install pCO₂ sensor

1.3 Lake Acidification & Invasive Carp

- Collect water sample

→ TMS

- Determine pH

→ C&C

- Determine if unknown sequences include invasive carp

→ C&C

- Model invasive carp

Task #2

2.1 Produce Power

- Retrieve power connector

- Install Power connector

- Replace sacrificial anode

- Mitigate corrosion

→ TMS

- Visual Inspection of structure → C&C

- Apply Patch to area

→ TMS

2.2 Monitor Environmental Impacts

- Collect polyp jellies → TMS
- Visually detect medusa jelly → C&C
- Collect Medusa Jelly → TMS
- Collect Fish species
- Remove Hydrophane
- Place Hydrophane
- pull pin

TMS

Final Result:

- Replace/Remove Cover
- Replace Thermistor
- Install pCO₂ sensor
- Collect water sample
- Retrieve/Install Power connector
- Replace Sacrificial Anode
- Apply patch to rust
- Collect polyps
- Collect Medusa Jelly
- Collect Fish species
- Remove/Place Hydrophane
- Pull Pin

Tools

- Claw
- Claw (1/2) stabilize top/thermistor
- Claw
- Claw w syringe?
- Holds metal hook
- Claw
- Wash cloth
- hook/handle size
- hinge
- balls
- handle (40 cm rope)
- pin size

Precision needed for claw

Task 1.1 $\frac{1}{2}$ inch PVC Pipe

Task 1.2 $\frac{1}{2}$ inch PVC

Task 1.3 $\frac{3}{4}$ inch

(Epoxy) Wash cloth ($12'' \times 12''$)

(Polyp) Pipe Cleaner

(Medusa Jelly) Water Wiggler ($9.61 \times 6.02 \times 2.9$) inches

(Fish) Ping pong balls

~~Plastic~~
(Pin) 9" length 1" hook 4mm diameter

#6 Crew Hook

$\frac{1}{2}$ inch pipe connector 1/4 inch